

CSE2430 - Operating Systems

Homework Security (5)

Assigned on: **Mar 06 2023**

1 Protection

Explain why a program would voluntarily switch from one domain to a domain with less capabilities, for example at function-calls.

2 Security

You are working for a company which develops communication software (e.g. like WhatsApp, Telegram, Facebook Messenger). Which security aspects do you take into consideration? Explain for each aspect why it is relevant.

3 Buffer Overflow

- 1) Explain what a buffer overflow exploit is and how it works.
- 2) Describe how buffer overflow exploits can be prevented at different levels: user, compiler, operating system.